

Q1 What is excess-3 codes? Please explain how to calculate excess-3 code from a BCD code. Draw circuit diagram of it.

Excess-3 code: Excess-3 (XS-3) code is a non-weighted binary-coded decimal (BCD) system where each decimal digit represented by a 4-bit binary code derived by adding 3 to its standard 8-4-2-1 BCD equivalent. It's also called the self-complementing code.

BCD to XS-3 code:

1. Write the BCD (8-4-2-1) code of the decimal digit.
2. Add 3 (0011) to that BCD code.
3. The result is the XS-3 code.

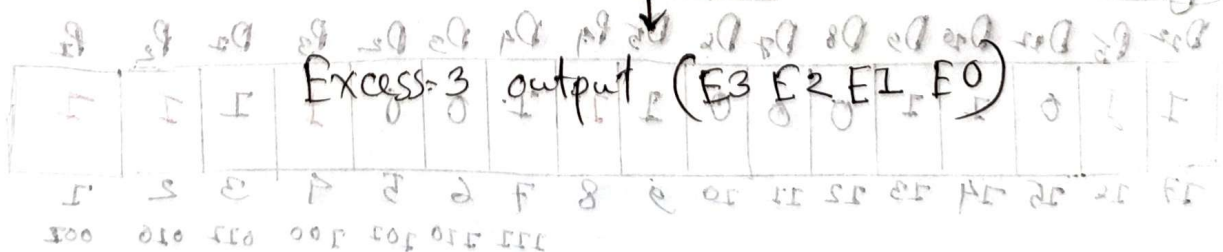
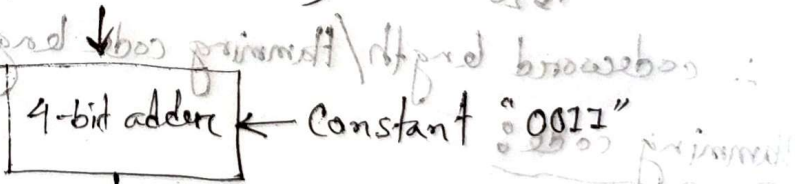
Examples:

Decimal	BCD code	+0011	XS-3 code
0	0000	+0011	0011
1	0001	+0011	0100

Circuit Diagram (BCD to XS-3 converter):

Next Photo/DSDNOTE

BCD Input (A3 A2 A1 A0)



Truth table: BCD to Excess-3 code

Decimal	BCD				Excess-3			
	A	B	C	D	W	X	Y	Z
0	0	0	0	0	0	0	1	1
1	0	0	0	1	0	1	0	0
2	0	0	1	0	0	1	0	1
3	0	0	1	1	0	1	1	0
4	0	1	0	0	0	1	1	1
5	0	1	0	1	1	0	0	0
6	0	1	1	0	1	0	0	1
7	0	1	1	1	1	0	1	0
8	1	0	0	0	1	0	1	1
9	1	0	0	1	1	1	0	0
10	1	0	1	0	X	X	X	X
11	1	0	1	1	X	X	X	X
12	1	1	0	0	X	X	X	X
13	1	1	0	1	X	X	X	X
14	1	1	1	0	X	X	X	X
15	1	1	1	1	X	X	X	X

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K-MAP - Simplified expression:

W \ AB \ CD	00	01	11	10
00				
01		1	1	1
11	X	X	X	X
10	1	1	X	X

$$W = A + BD + BC$$

$$= A + B(C + D)$$

X \ AB \ CD	00	01	11	10
00		1	1	1
01	1			
11	X	X	X	X
10		1	X	X

$$X = BC'D' + B'C + B'D$$

$$= BC'D' + B'(C + D)$$

$$= B(C + D)' + B'(C + D)$$

$$= B \oplus (C + D)$$

Y \ AB \ CD	00	01	11	10
00	1		1	
01	1		1	X
11	X	X	X	X
10	1		X	X

$$Y = C'D' + CD$$

$$= (C \oplus D)'$$

$$= C \odot D$$

Z \ AB \ CD	00	01	11	10
00	1			1
01	1			1
11	X	X	X	X
10	1		X	X

$$Z = C'D' + CD'$$

$$= D'(C + C')$$

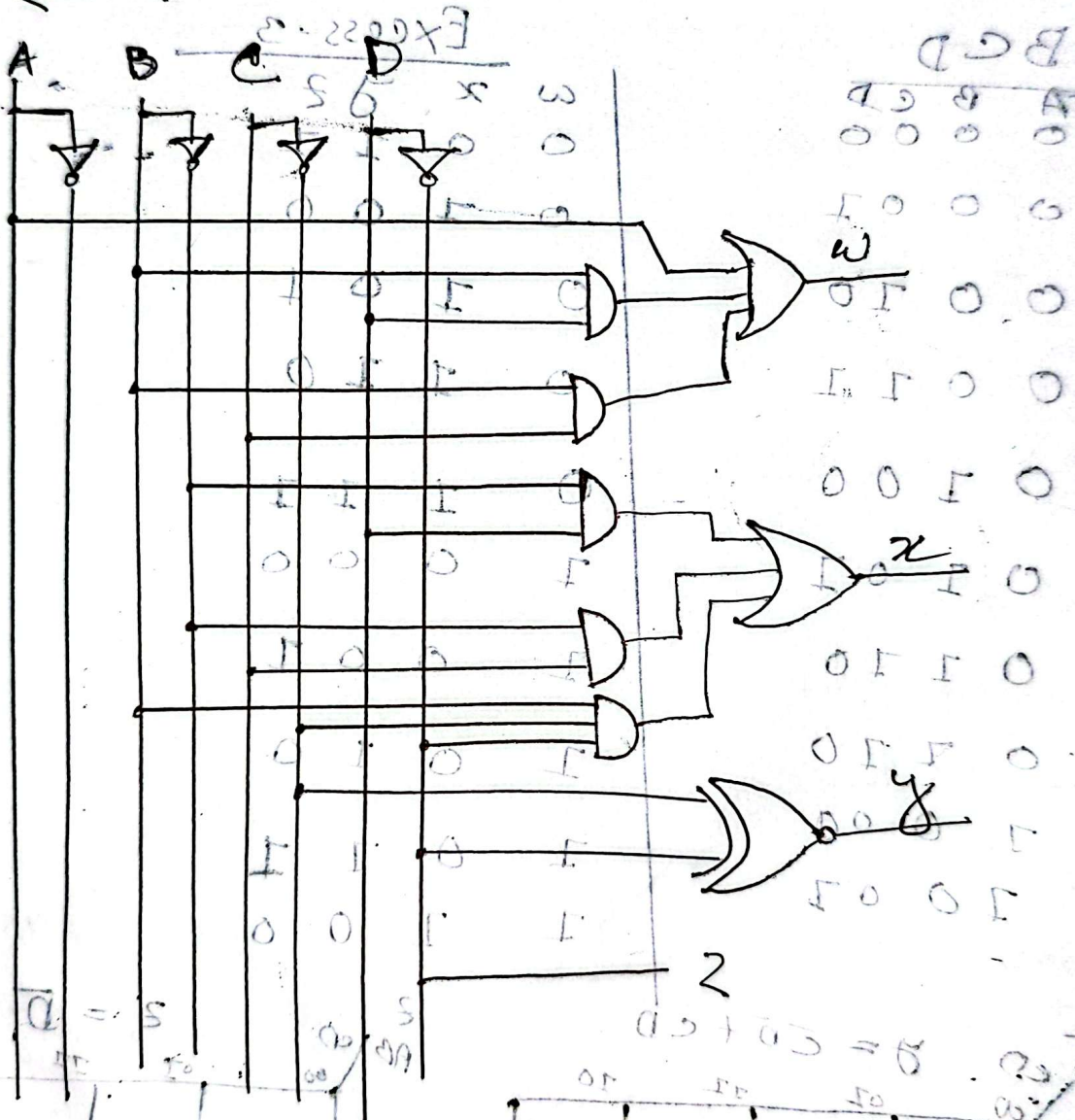
$$= D'$$

$$W = A + BD + BC$$

$$X = \bar{B}D + \bar{B}C + B\bar{C}\bar{D}$$

$$Y = \bar{C}\bar{D} + CD \quad (C \oplus D)$$

$$Z = \bar{D}$$



1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0

2. Compute the Binary data word with Hamming code for the Binary data word $(101100011001)_2$ or $(2020-5)$

Ans: Given, $(101100011001)_2$
We know,

$$2^r \geq m + r + 1$$

where,

$$m \text{ (message bit)} = 12$$

$$r \text{ (required Parity bit)} = ?$$

if, $r = 5$

then, $2^5 \geq 12 + 5 + 1$

$$\Rightarrow 32 \geq 18$$

So, $r = 5$ parity bits are required

Positioning the parity bits:

Binary	BCD code
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
10	1010
11	1011
12	1100
13	1101
14	1110
15	1111

$$P_{2^1} = 2^0 = 1 \quad / P_1$$

$$P_{2^2} = 2^1 = 2 \quad / P_2$$

$$P_{2^3} = 2^2 = 4 \quad / P_4$$

$$P_{2^4} = 2^3 = 8 \quad / P_8$$

$$P_{2^5} = 2^4 = 16 \quad / P_{16}$$

Total length

$$\therefore \text{codeword length/Hamming code length} = m + r = 12 + 5 = 17 \text{ bit}$$

Hamming code:

D ₁₂	P ₅	D ₁₁	D ₁₀	D ₉	D ₈	D ₇	D ₆	D ₅	P ₄	D ₄	D ₃	D ₂	P ₃	D ₁	P ₂	P ₁
1	1	0	1	1	0	0	0	1	1	1	0	0	1	1	1	1
17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
										111	110	101	100	011	010	001

$$P_1 = \text{X-OR of bits } (D_1, D_2, D_4, D_5, D_7, D_9, D_{11}, D_{12})$$

$$= 1 \oplus 0 \oplus 1 \oplus 1 \oplus 0 \oplus 1 \oplus 0 \oplus 1 = 1$$

$$P_2 = \text{X-OR of bits } (D_1, D_3, D_4, D_6, D_7, D_{10}, D_{11})$$

$$= 1 \oplus 0 \oplus 1 \oplus 0 \oplus 0 \oplus 1 \oplus 0 = 1$$

$$P_3 = \text{X-OR of bits } (D_2, D_3, D_4, D_8, D_9, D_{10}, D_{11})$$

$$= 0 \oplus 0 \oplus 1 \oplus 0 \oplus 1 \oplus 1 \oplus 0 = 1$$

$$P_4 = \text{X-OR of bits } (D_5, D_6, D_7, D_8, D_9, D_{10}, D_{11})$$

$$= 1 \oplus 0 \oplus 0 \oplus 0 \oplus 1 \oplus 1 \oplus 0 = 1$$

$$P_5 = \text{X-OR of bits } (D_{12})$$

$$= 1 \oplus 0 = 1$$

D_{17}	P_{16}	D_{15}	D_{14}	D_{13}	D_{12}	D_{11}	D_{10}	D_9	P_8	D_7	D_6	D_5	P_4	D_3	P_2	P_1
1	1	0	1	1	0	0	0	1	1	1	0	0	1	1	1	1

What is Priority Encoder? Draw 4-to-2 Priority Encoder.

Ans: A priority encoder is a type of encoder that assigns priority to inputs.

If two or more inputs are 1 at the same time, the encoder outputs the binary code of the highest-priority input.
e.g., $D_3 > D_2 > D_1 > D_0$.

4-to-2 Priority Encoder:

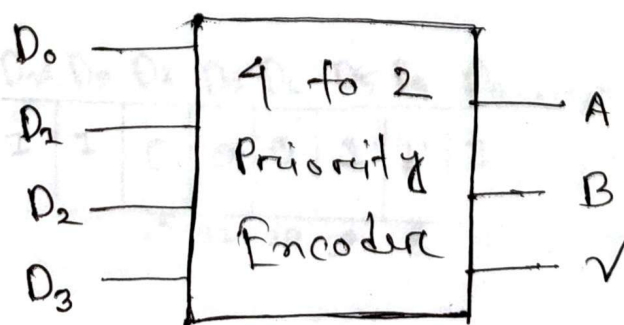
Truth table:

Input				Output		Validity
D_3	D_2	D_1	D_0	Y_1	Y_0	$\checkmark$
0	0	0	0	0	0	0
0	0	0	1	0	0	1
0	0	1	X	0	1	1
0	1	X	X	1	0	1
1	X	X	X	1	1	1

$\checkmark = 0$ (invalid)

$\checkmark = 1$ (valid)

Block diagram: (প্রাইরিতীর এন্কোডার)



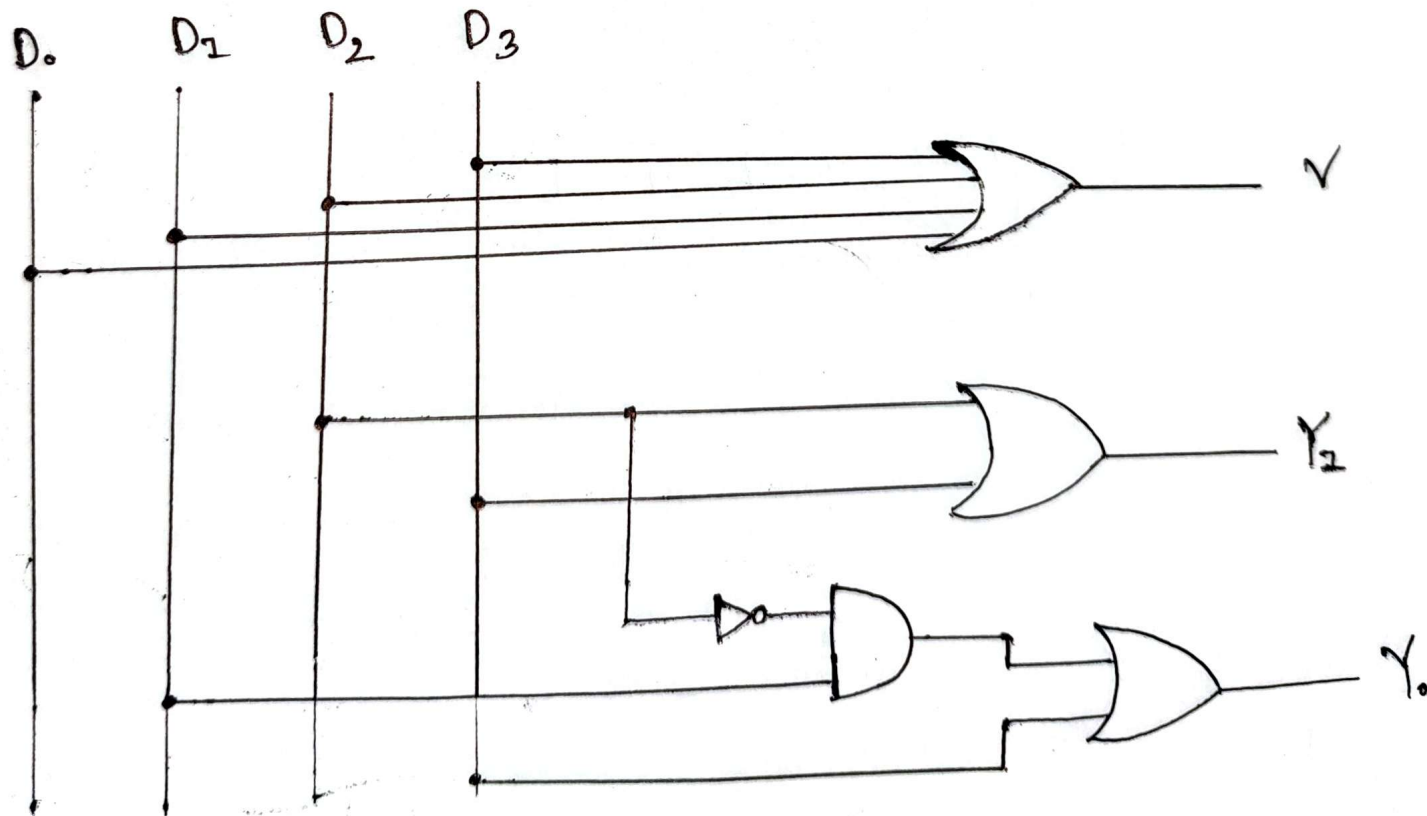
from truth table -

$$v = D_0 + D_1 + D_2 + D_3$$

$$Y_0 = \cancel{D_1} D_3 + \bar{D}_3 \bar{D}_2 D_1 = (D_3 + \bar{D}_3)(D_3 + \bar{D}_2 D_1) = D_3 + \bar{D}_2 D_1$$

$$Y_1 = D_3 + \bar{D}_3 D_2 = (D_3 + \bar{D}_3)(D_3 + D_2) = D_3 + D_2$$

logic circuit diagram:



5. Design a Full subtractor circuit with a decoder & two OR gates.

Ans: The truth table for the circuit is given below.

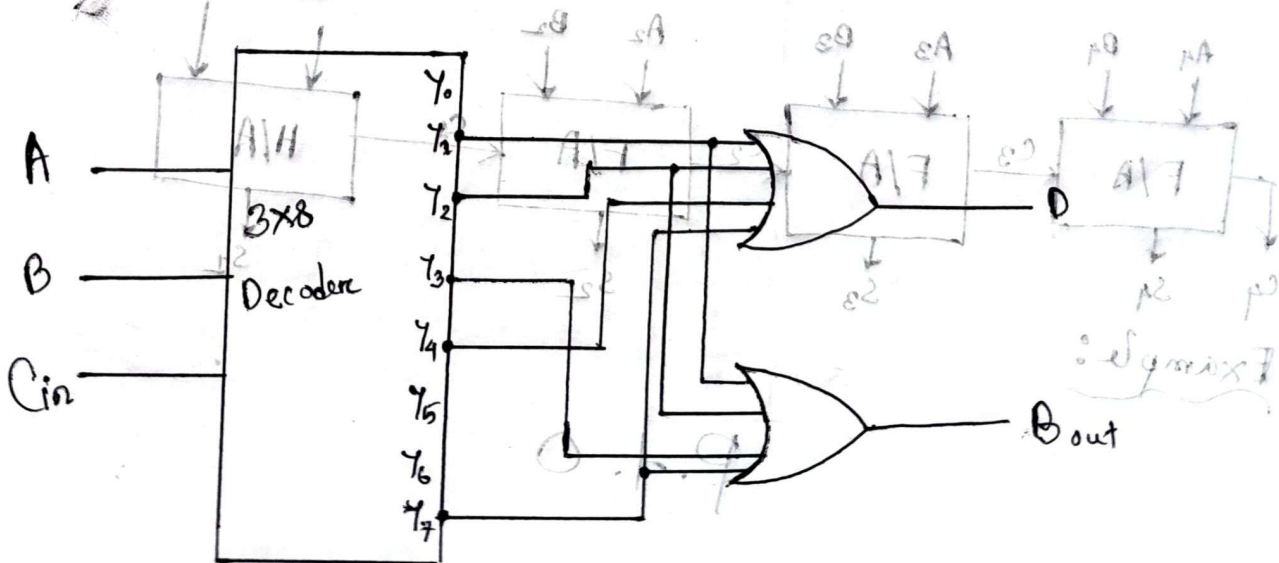
Truth table

A	B	C _{in}	Difference (D)	Borrow (B _{out})
0	0	0	0	0
0	0	1	1	1
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

$$\therefore D = \sum m(1, 3, 4, 7)$$

$$B_{out} = \sum m(1, 2, 3, 7)$$

Circuit diagram: (Using 3x8 decoder & two OR gates)



6. Compare between ripple carry adder and carry look-ahead adder.

Answer:

Compare between RCA & CLA :

Aspects	RCA	CLA
Structure	Series connection of full adders.	Uses generate (G) and propagate (P) logic
Carry calculation	Carry ripples from one stage to the next.	Carry is computed in advance.
Speed	Slow.	Fast
Complexity	Simple design	Complex design
Power	Lower	Higher
Use case	Small bit size addition	Larger addition

7. Design a logic circuit that has three inputs, A, B, and C and whose output will be High only when a majority of the inputs are High.

Answer:

truth table:

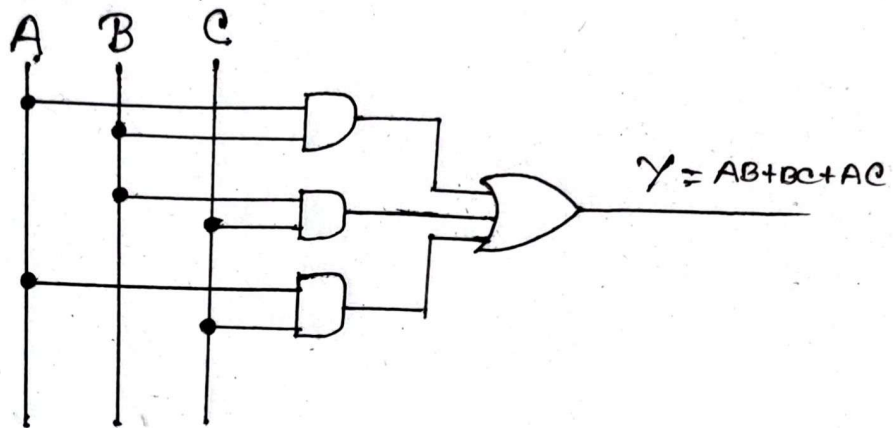
Input			Output
A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$\therefore$ ~~map~~

Y \ BC				
	00	01	11	10
A				
0			1	
1		1	1	1

$$\therefore Y = AB + BC + AC$$

Circuit diagram:

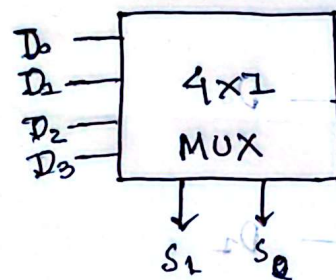


9. Define Multiplexer. Design a 4-bit input MUX. How to design an 8-input MUX using two 4-input MUX.

Ans: Multiplexer is a combinational logic circuit that used to select only one input among several inputs based on selection lines.

- This can be act as digital switch.
- This is also called the data selector.
- For a MUX There can be 2^n inputs, n selection line & only one output

Block diagram: (4x1) MUX

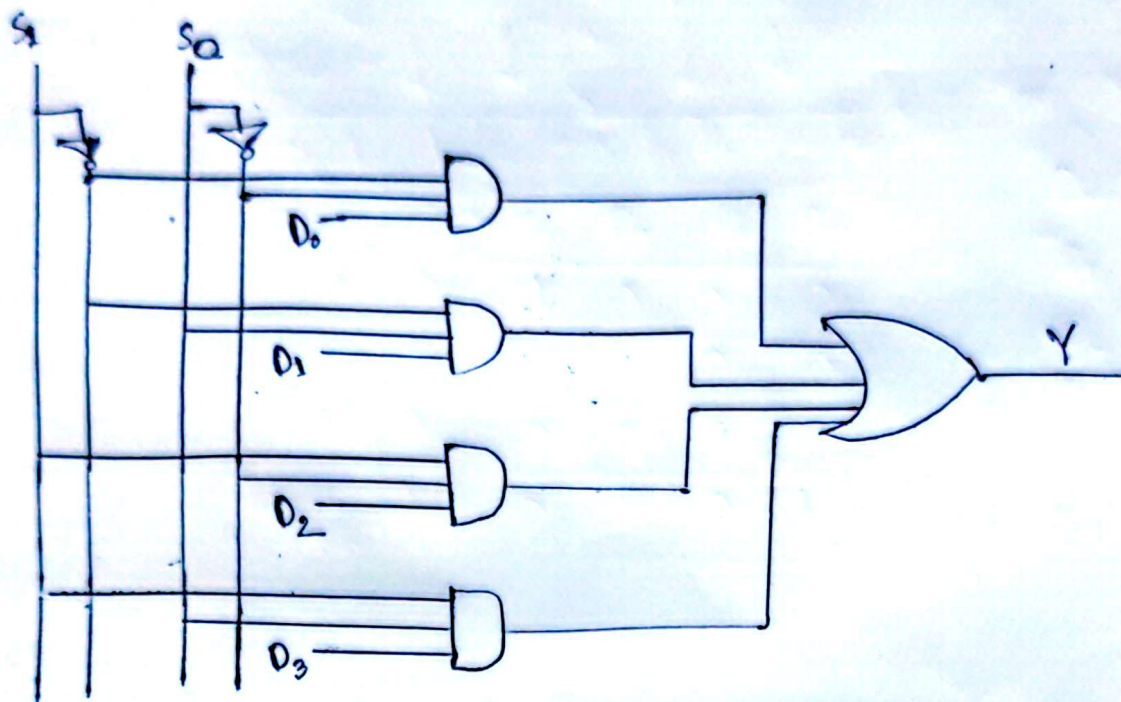


Truth table:

S_1	S_0	Y
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

$$\therefore Y = \bar{S}_1 \bar{S}_0 D_0 + \bar{S}_1 S_0 D_1 + S_1 \bar{S}_0 D_2 + S_1 S_0 D_3$$

logic circuit:

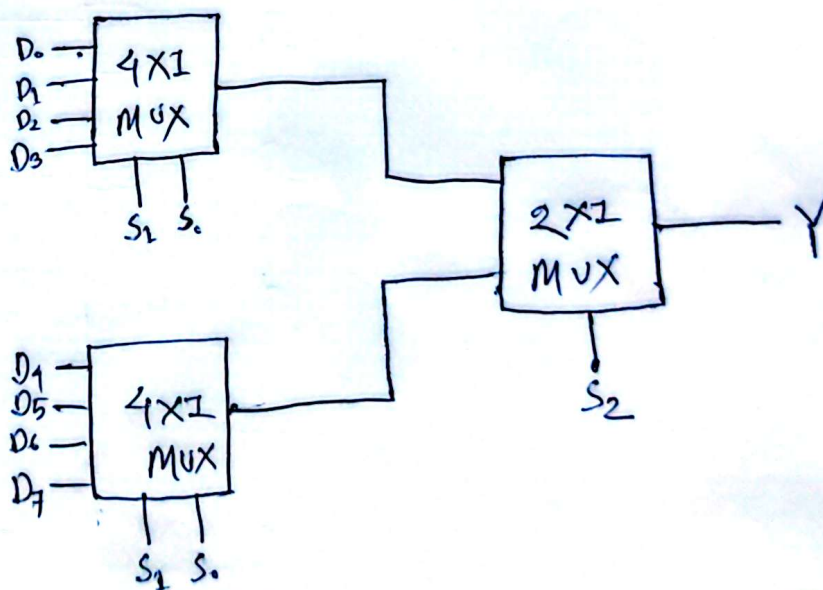


Design of an 8-input MUX using two 4-input MUX:

Truth table:

S_2	S_1	S_0	Y
0	0	0	D_0
0	0	1	D_1
0	1	0	D_2
0	1	1	D_3
1	0	0	D_4
1	0	1	D_5
1	1	0	D_6
1	1	1	D_7

diagram:



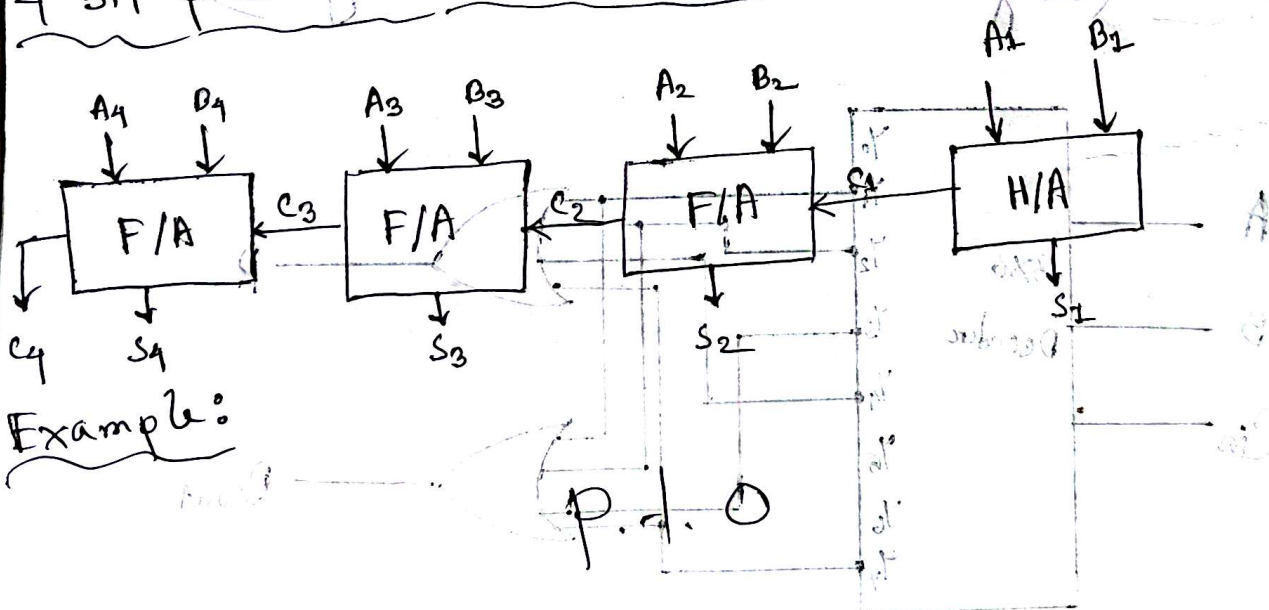
10. What are the advantages of Parallel subtractor? Design a 4-bit parallel adder. Also evaluate the circuit with an example.

Ans: Advantages of Parallel subtractor: A digital circuit that is used to calculate the arithmetic difference between two binary pairs of bits is known as a parallel subtractor.

Advantages:

1. The operation of this subtractor is faster when constructed to serial subtractor.
2. The required time for subtraction doesn't depend on the digit of bits.
3. All the bits in this are subtracted at a time, so the output will be in parallel form.
4. It not expensive.
5. This are faster compared with serial counterparts.

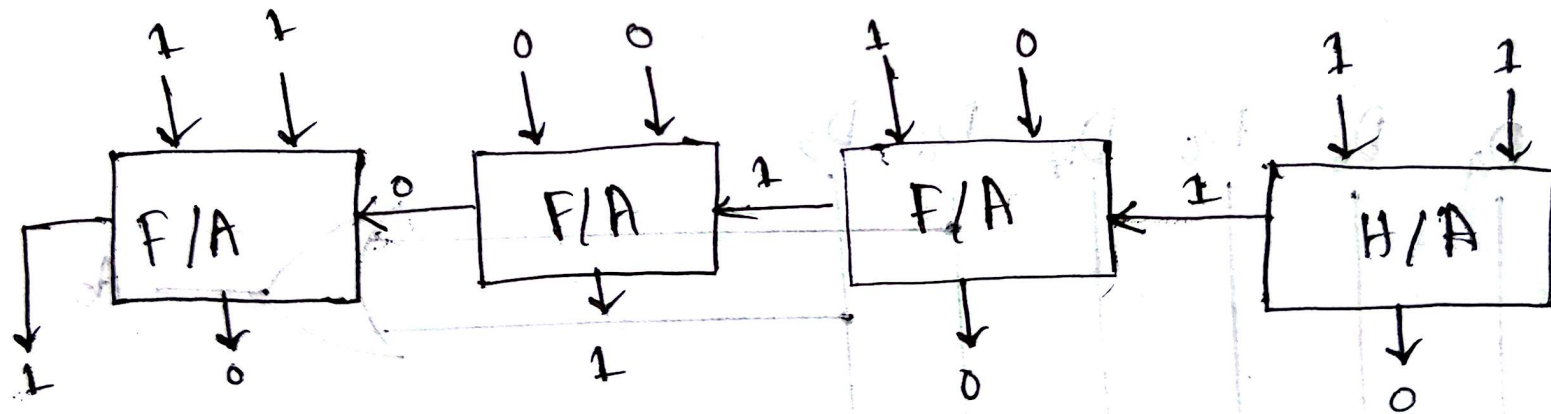
4 bit parallel adder Diagram:



Example:

P. 1. 0

For adding $(1011)_2$ & $(1001)_2$



$$\therefore (1011)_2 + (1001)_2 = 10100$$

$$\begin{array}{r} 1011 \\ 1001 \\ \hline 10100 \end{array}$$

11. Advantages and disadvantages of magnitude comparator :

Advantages	Disadvantages
performs fast as to compare two binary numbers	complexity increases when bits increases.
Simple design for small bit comparator	Requires more logic gates for higher bits
provides 3 useful outputs like $A > B$, $A = B$, $A < B$	Comparators can't perform arithmetic
Useful in CPU, memory, calculators	works only comparison
Saves time compared to manual check	less space efficient

12. what is magnitude comparator? Design a 2-bit magnitude comparator with truth table and derive the logical formula.

A magnitude comparator is a combinational logic circuit that compares two binary numbers that indicates one number is greater, less or equal to the other one.

truth table:

Input				Output		
A ₁	A ₀	B ₁	B ₀	A > B	A = B	A < B
0	0	0	0	0	1	0
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	0	1
0	1	0	0	1	0	0
0	1	0	1	0	1	0
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	1	0	0
1	0	0	1	1	0	0
1	0	1	0	0	1	0
1	0	1	1	0	0	1
1	1	0	0	1	0	0
1	1	0	1	1	0	0
1	1	1	0	1	0	0
1	1	1	1	0	1	0

Let, $G = A > B$, $E = (A = B)$, $L = A < B$

$\therefore G$

$A_1 A_0$ (rows: 00, 01, 11, 10)
 $B_1 B_0$ (columns: 00, 01, 11, 10)

Handwritten 1s in the following cells: (01, 00), (01, 01), (11, 01), (11, 10), (10, 01), (10, 10).

$$G = A_1 \bar{B}_1 + A_0 \bar{B}_1 \bar{B}_0 + A_1 A_0 \bar{B}_0$$

E

$A_1 A_0$ (rows: 00, 01, 11, 10)
 $B_1 B_0$ (columns: 00, 01, 11, 10)

Handwritten 1s in the following cells: (00, 00), (01, 01), (11, 11), (10, 10).

$$E = \bar{A}_1 \bar{A}_0 \bar{B}_1 \bar{B}_0 + \bar{A}_1 \bar{A}_0 \bar{B}_1 B_0 + A_1 A_0 B_1 B_0 + A_1 A_0 B_1 \bar{B}_0$$

$$= \bar{A}_1 \bar{B}_1 (\bar{A}_0 \bar{B}_0 + A_0 B_0) + A_1 B_1 (B_0 A_0 + \bar{A}_0 \bar{B}_0)$$

$$= \overline{A_0 \oplus B_0} \cdot \overline{A_1 \oplus B_1}$$

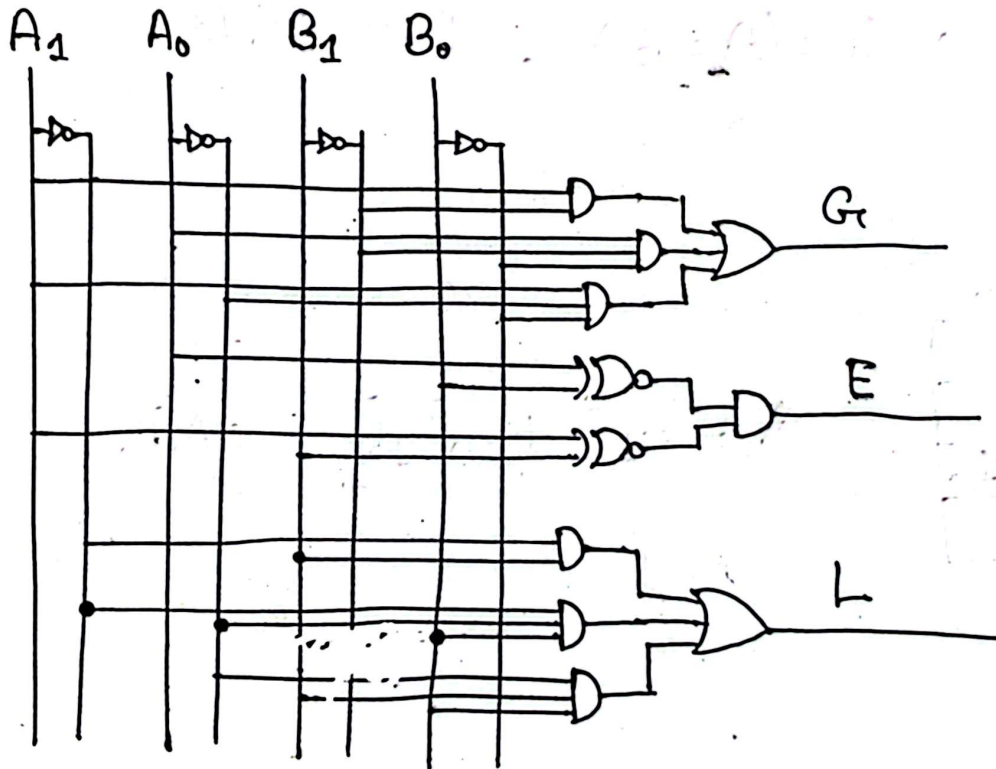
L

$A_1 A_0$ (rows: 00, 01, 11, 10)
 $B_1 B_0$ (columns: 00, 01, 11, 10)

Handwritten 1s in the following cells: (01, 01), (01, 11), (11, 01), (11, 10), (10, 11).

$$L = \bar{A}_1 B_1 + \bar{A}_1 \bar{A}_0 B_0 + B_1 B_0 \bar{A}_0$$

circuit diagram!



13. Design a decimal to BCD encoder with circuit diagram.

Inputs: 10

Outputs: 4

Truth table:

Decimal Inputs		BCD outputs			
		A ₃	A ₂	A ₁	A ₀
D ₀	0	0	0	0	0
D ₁	1	0	0	0	1
D ₂	2	0	0	1	0
D ₃	3	0	0	1	1
D ₄	4	0	1	0	0
D ₅	5	0	1	0	1
D ₆	6	0	1	1	0
D ₇	7	0	1	1	1
D ₈	8	1	0	0	0
D ₉	9	1	0	0	1

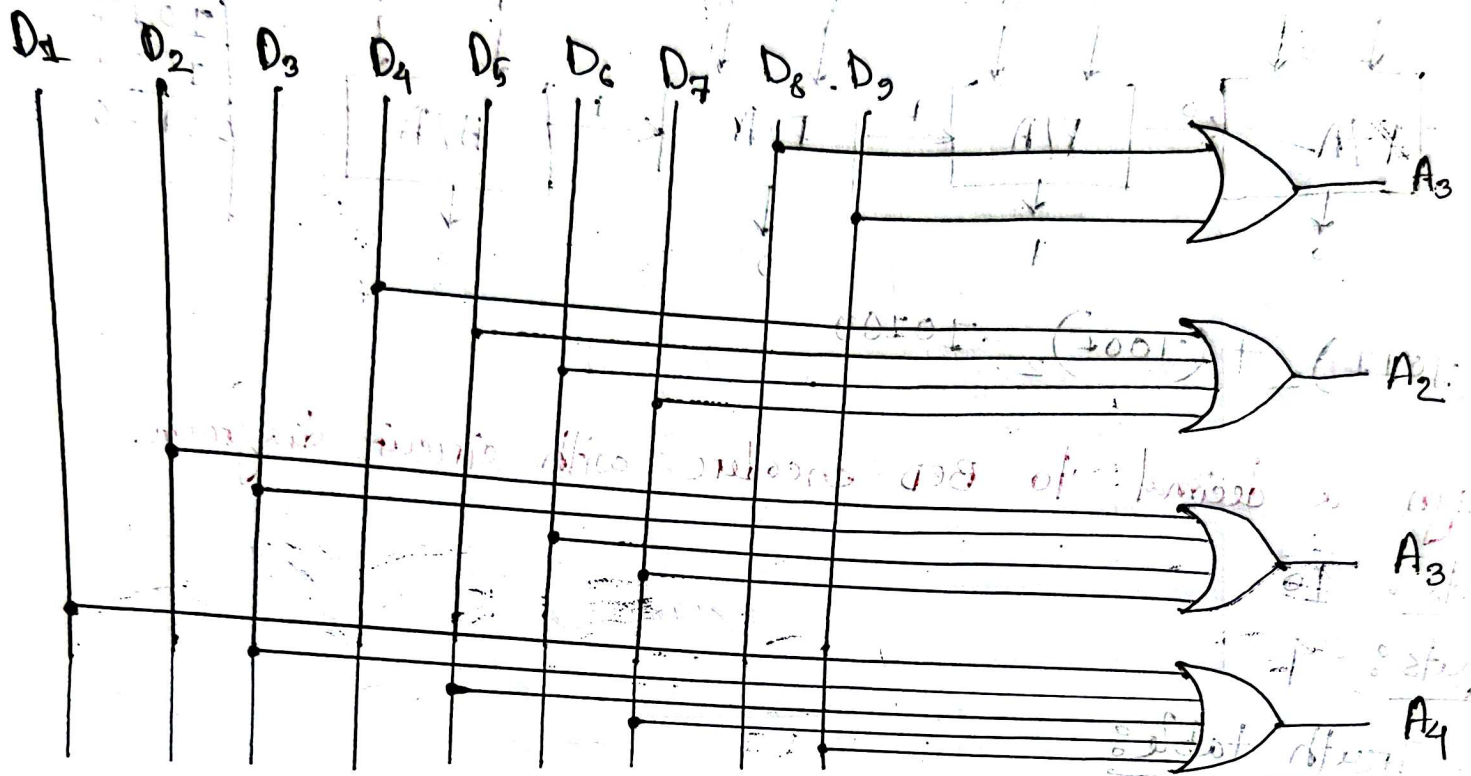
$$A_3 = D_8 + D_9$$

$$A_2 = D_4 + D_5 + D_6 + D_7$$

$$A_1 = D_2 + D_3 + D_6 + D_7$$

$$A_0 = D_1 + D_3 + D_5 + D_7 + D_9$$

Circuit diagram:



14. Write the application of Encoders. Design a 2-to-4 binary decoder.

Application of Encoders:

1. Computer System: Keyboards to binary code.
2. Motor speed. Sync: Shaft speed/Position feedback.
3. CNC Machines: Accurate tool positioning.
4. Industry: Automation & robotics control.
5. Medical: CT scanners, surgical devices.
6. Data Routing: Input lines to binary code.
7. Signal Processing: Data compression & transmission.

2-to-4 binary decoder:

Truth table:

Input (A, B)		Output (D ₀ , D ₁ , D ₂ , D ₃)			
A	B	D ₀	D ₁	D ₂	D ₃
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

Minterm Equation:

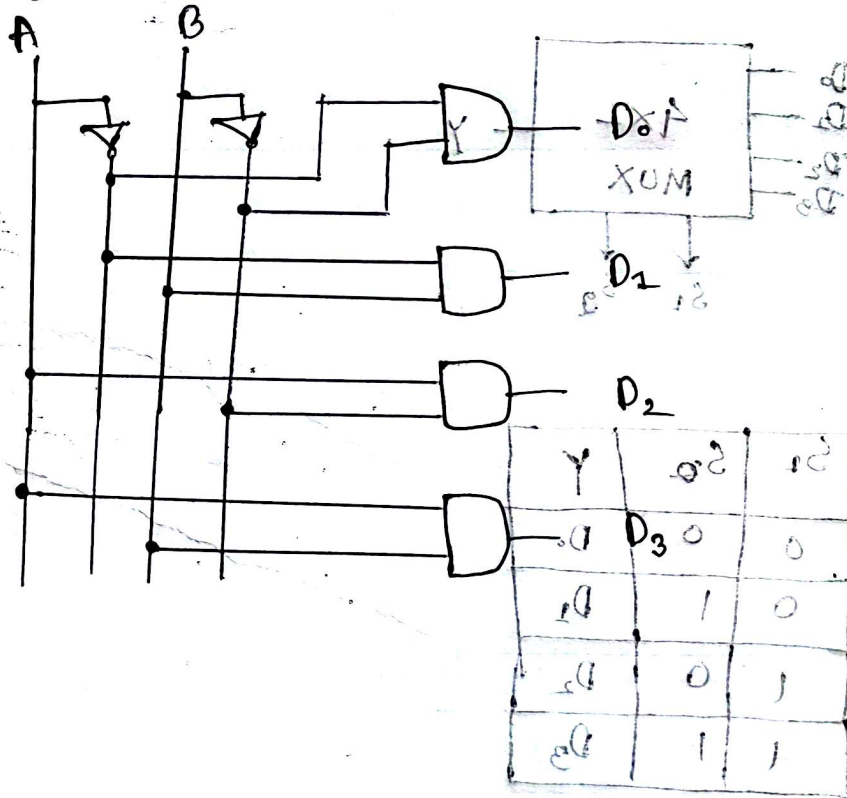
$$D_0 = \bar{A} \bar{B}$$

$$D_1 = \bar{A} B$$

$$D_2 = A \bar{B}$$

$$D_3 = A B$$

Logic diagram:



$$D_0 = \bar{A} \bar{B} + \bar{A} B + A \bar{B} + A B = 1$$

Note: $\left[\begin{array}{l} \text{odd} \rightarrow d \rightarrow 1 \\ \text{even} \rightarrow \bar{d} \rightarrow 0 \end{array} \right] D \rightarrow \text{data line}$

15. Design the following circuit using

MUX: $F(A, B, C, D) = \sum (1, 3, 5, 6, 8, 11, 13, 15)$

truth table:

Input					Out
No.	A	B	C	D	F
0	0	0	0	0	0
1	0	0	0	1	1
2	0	0	1	0	0
3	0	0	1	1	1
4	0	1	0	0	0
5	0	1	0	1	1
6	0	1	1	0	1
7	0	1	1	1	0
8	1	0	0	0	1
9	1	0	0	1	0
10	1	0	1	0	0
11	1	0	1	1	1
12	1	1	0	0	0
13	1	1	0	1	1
14	1	1	1	0	0
15	1	1	1	1	1

Implementation table:

	$\bar{D}$	D
D ₀	0	①
D ₁	2	③
D ₂	4	⑤
D ₃	⑥	7
D ₄	⑧	9
D ₅	10	⑪
D ₆	12	⑬
D ₇	14	⑮

here, D₀ = D

D₁ = D

D₂ = D

D₃ = $\bar{D}$

D₄ = $\bar{D}$

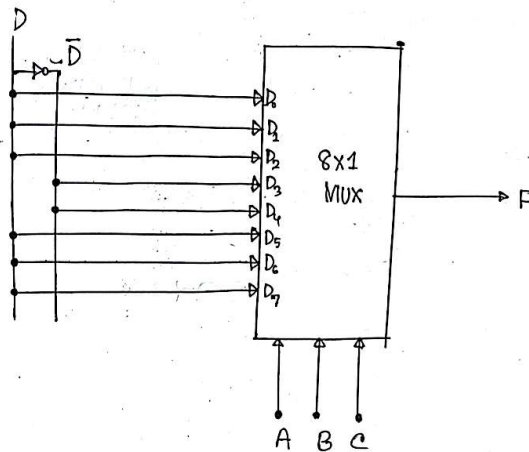
D₅ = D

D₆ = D

D₇ = D

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circuit diagram:



here, D = logic high

$\bar{D}$ = logic low

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করা

15.

Implementation table:

	$\bar{c}\bar{d}$	$\bar{c}d$	$c\bar{d}$	cd
D_0	0	1	2	3
D_1	4	5	6	7
D_2	8	9	10	11
D_3	12	13	14	15

Note

১০মঃ: এর Ans (4x1)
mux দিয়ে করা যাবে।
কিউ চাইলে এটা দিয়েও
করাতে পারবে।

১০মঃ: $cd \rightarrow$ মেথান
মেথান আছে (C,D)
হবে।

$$D_0 = \bar{c}\bar{d} + \bar{c}d = \bar{c}(d + \bar{d}) = \bar{c}$$

$$D_1 = \bar{c}d + c\bar{d} = c \oplus d$$

$$D_2 = \bar{c}\bar{d} + cd = \overline{(c \oplus d)} = c \odot d$$

$$D_3 = \bar{c}d + cd = c$$

